

Arash Daneshvar

Robotics & AI Software Engineer

with hands-on experience deploying deep learning models on NVIDIA Edge platforms for real-world robotic manipulation, including work on Unitree G1 (humanoid) and ABB GoFa (manipulation arm) robot. Skilled in bridging learning-based approaches (RL, IL, VLMs) with advanced robotic systems for simulation and industrial validation.

✉ ao.daneshvar@gmail.com

📍 Torino, Italy

in [LinkedIn](#)

☎ +39 379 232 4690

🐙 [GitHub](#)

🌐 [Website](#)

Work Experience

MakrShakr - Turin, Italy | Mar 2025 - Now

"Developing Intelligent Perception, Imitation Learning, and Robotic Manipulation Solutions for Bartender Robots"

- **Imitation Learning for Humanoid Robot:** Collaborated with the VANDAL Lab (PoliTo) to develop Imitation Learning and Behavior Cloning frameworks for humanoid robot control on Robi, humanoid robot built on the Unitree G1, supporting the transition from simulation testing to physical hardware validation.
- **Real-Time Object Detection:** Deployed high-performance, optimized object detection models on NVIDIA Jetson Edge AI platforms, achieving a 94% mAP@50-95 for challenging, highly reflective, and transparent objects.
- **Autonomous Pick and Place:** Integrated vision-based perception with 3D spatial grounding to execute low-latency pick-and-place tasks using an ABB GoFa arm, utilizing Hand-Eye calibration and ABB EGM communication protocols.

Projects

Robot Learning & Advanced Simulation - *"Domain Randomization in Reinforcement Learning: Sim-to-Real Transfer"*

- Developed reinforcement learning-based control policies for robotics, focusing on sim-to-real transfer and domain randomization. **Tools:** OpenAI Gym, PyTorch, Python | [GitHub](#)

ML for IoT - *"Smart Battery Monitoring with Voice Processing"*

- Developed an IoT-based system for remote battery monitoring with voice control and real-time visualization, applying optimization algorithms to enhance battery efficiency and system performance. **Tools:** TensorFlow Lite, Redis, Python | [GitHub](#)

Computer Vision - *"Image Geo-localization Through Retrieval"*

- Built a deep learning-based image geo-localization system that predicts candidate city locations from input images using visual feature retrieval. **Tools:** PyTorch, Python | [GitHub](#)

LLM - *"Enhanced RAG Retrieval to Expand LLM Context"*

- Developed a retrieval method for RAG architectures, improving context relevance and retrieval completeness for LLM-based systems on MS MARCO dataset. **Tools:** Python, FAISS, LLMs (Groq API) | [GitHub](#)

Education

MSc | Data Science and Engineering

Politecnico di Torino | Sep 2022 - Mar 2026

Deep learning | Big data | ML for IoT | Advanced Robotics | Robot Learning

MSc | Robotics Engineering

Amirkabir University | Sep 2015 - Nov 2018

Non-linear Control | Adaptive Control | Optimization

BSc | Robotics Engineering

Shahrood University | Sep 2011 - Jul 2015

Robotics | Dynamics | Electric circuit | Electronics | Programming

Technical Skills

Robotics: ROS/ROS2, EGM, RAPID, Motion control, Gazebo, NVIDIA Isaac Sim (Omni-verse)

AI: Real-time Computer Vision, LLM, VLM, VLA, RAG

Embedded Systems: NVIDIA Jetson series, Raspberry Pi, Edge AI

Hardware: Unitree G1, ABB GoFa Arm robot, Intel RealSense D435 camera

Programming & Software: Python, C++, Docker, git, Linux

Research Interests

- Robot Learning (RL | IL | CL)
- Computer Vision & VLM & LLM
- IoT and Embedded systems

Languages

- Persian (Native)
- English (C1)
- Italian (B1)